

AI GreetCraft

*Figma plugin for postcards creating using GenAI technologies

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I. INTRODUCTION

Greeting cards are a timeless means of conveying emotions during special occasions. However, the process of creating visually appealing and heartfelt cards often requires design skills and creative writing abilities. With recent advances in generative AI, it is now possible to automate this creative process while preserving a personalized touch. AI GreetCraft is an AI-powered Figma plugin designed to generate personalized greeting cards by combining layout generation, user-uploaded photos, a custom AI-generated poem and a thematic AI-generated background. The plugin enables users to create aesthetically pleasing and emotionally resonant cards directly within Figma, using a seamless and intuitive interface powered by GenAI technologies.

II. RELATED WORK

Several AI-driven tools have emerged to automate creative content generation, particularly for visual and textual artifacts. For example, AI Card Craft is a web application that combines AI-generated artwork with simple greeting messages. However, such tools typically treat text, image, and layout as separate, loosely connected stages.

In contrast, AI GreetCraft offers a fully integrated card creation pipeline, from personalized poem generation using large language models (LLMs) to automated layout generation and image synthesis - all embedded directly within the Figma design environment. This end-to-end integration allows for a more coherent and customizable user experience.

Previous research has demonstrated the efficacy of LLM in creative text generation tasks. For example, [1] presents a case of fine-tuning the Gemma 2 model for Bengali poetry generation, highlighting the adaptability of LLMs to poetic structures and cultural nuances. On the visual side, the Hugging Face Diffusers library [2] enables the training of custom text-to-image models, facilitating thematic coherence between text and imagery.

Furthermore, the Figma Plugin API [3] has opened the possibilities to integrate generative workflows directly into professional design tools. While prior tools and studies focus on isolated generative capabilities, AI GreetCraft demonstrates how a unified generative approach can streamline the creative process and enhance personalization at scale.

Recent studies have also explored the use of generative models as creative amplifiers. For example, [4] discusses

how LLMs can be used not just for content generation but for ideation, variation, and stylistic experimentation. This aligns with our use of prompt-based keyword conditioning to generate emotionally resonant poetry and visuals that feel both novel and personalized.

These contributions underscore the growing potential of generative AI to produce emotionally resonant, user-tailored content in both textual and visual domains.

III. METHODOLOGY

A. Overview

Our system consists of three main components:

- 1) **Poem Generator** – a fine-tuned LLM for English occasion-specific poetry.
- 2) **Background Generator** – a custom-trained diffusion model for thematic background generation.
- 3) **Figma Layout Composer** – a plugin utilizing the Figma API to arrange visual elements dynamically.

B. Poem Generation Model

To generate custom poems in English, we fine-tuned a series of transformer models using parameter-efficient fine-tuning (PEFT) and Low-Rank Adaptation (LoRA). The training dataset consisted of 2,034 curated samples, each containing a *Title*, *Theme*, and *Poem*, parsed from a publicly available poetry website.

We experimented with several model architectures, including T5-small, GPT2-medium, LLaMA3-1.8B, and Gemma 2B. These were chosen to explore trade-offs between model size, training speed, memory usage, and poem quality.

Our experiments were conducted using the **Unsloth ecosystem**, which enables efficient training on consumer-grade hardware through the following optimizations:

- **QLoRA** — for memory-efficient fine-tuning using low-rank adapters with quantized weights;
- **Flash Attention 2** — for faster attention computation during training;
- **8-bit and 4-bit quantization** — for reduced memory footprint without significant quality loss;
- **Unsloth** — a lightweight training framework optimized for fast and efficient PEFT fine-tuning on consumer GPUs.

The final system accepts prompts in the form of keywords, such as occasion, recipient's name, and mood descriptors.

These are used to generate original poems with stylistic coherence and emotional tone.

C. Background Image Generation Model

To generate high-quality occasion-themed background images, we fine-tuned a latent diffusion model — **Stable Diffusion (CompVis/stable-diffusion-v1-4)** — using random 2,000 samples from the **Describable Textures Dataset (DTD)**, which contains 5,640 diverse, texture-rich images well-suited for postcard backgrounds.

The training was conducted using the **Hugging Face Diffusers** library.

- **Image Size:** 256×256
- **Batch Size:** 4
- **Epochs:** 15
- **Learning Rate:** 1×10^{-4}
- **Optimizer:** 8-bit Adam optimizer (memory-efficient)
- **Scheduler:** Cosine learning rate scheduler with warm-up steps

Libraries and Tools:

- **Frameworks:** PyTorch, Hugging Face Diffusers
- **Libraries:** torchvision, datasets, bitsandbytes

The resulting model was pushed to the Hugging Face Hub: <https://huggingface.co/Nazgulitos/background-generator-with-stable-diffusion-v1-4>

D. Figma Plugin and Layout Design

The Figma Plugin API was employed to develop a user interface within Figma where users can:

- Upload a personal photo
- Input recipient name and select the occasion
- Automatically receive a full greeting card layout including the custom poem, generated background, and uploaded photo
- Optionally adjust the postcard elements and add new once, unlike if the postcard was generated as a single image

Each card layout is composed of four distinct visual elements: text, header, image, and background, all placed as separate objects on the Figma canvas. These objects are arranged using a placement algorithm that dynamically resizes and positions them based on:

- Figma canvas constraints
- Typography heuristics
- Color palette matching (based on background colors)

The visual parameters — such as font size, alignment, padding, and layering — are defined in an instruction JSON, which is generated by the GPT 4o and then safely parsed by the plugin. In parallel, the content for the postcard is generated by our fine-tuned models.

Each object can be anchored to one of nine predefined zones on the postcard grid (top left, top center, top right, middle left, etc.). If two or more elements are assigned to the same zone, they are stacked vertically as a column within that space, preserving visual hierarchy and readability.

E. Workflow Diagram

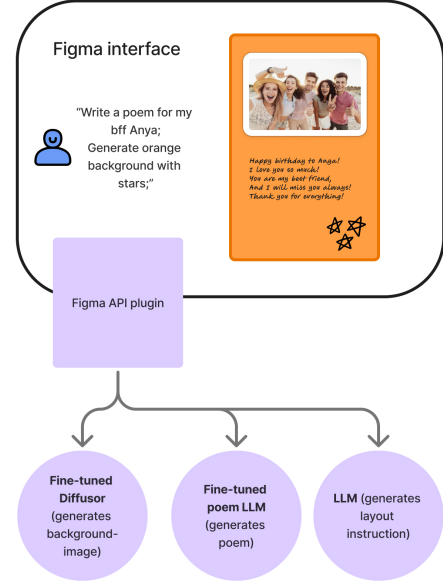


Fig. 1. Workflow Diagram

F. GitHub Link

<https://github.com/andrewlevada/postcard-generator-experiment>

G. Figma Plugin Link

<https://www.figma.com/community/plugin/1479114006154342367/postcard-generator-experiment>

IV. DATASET

Our system employs two distinct datasets: one for poem generation and another for background image synthesis.

A. Poem Generation Dataset

Source: Poems were collected from <https://www.wishafriend.com/poems/> using a custom web scraper (see [6]).

Structure: The dataset comprises 2,034 English poems, each annotated with:

- **Title:** A poetic descriptor (e.g., “Autumn’s Sorrow”).
- **Theme:** Occasion or tone (e.g., “Wedding,” “Romantic”).
- **Poem:** 4–8 rhymed lines (median length: 52 words).

Preprocessing:

- Normalized encoding (UTF-8) and whitespace.
- Reformatted entries into a prompt-based structure for conditional generation, combining the poem’s theme and title. The resulting template followed this instruction: “Below is a theme and title for a poem. Write a greeting poem that matches the theme and title of a greeting card.” This was followed by clearly labeled sections for *Theme*, *Title*, and *Poem*.
- Data were splitted into 90% training and 10% validation.

B. Background Image Dataset

Source: A curated subset of the Describable Textures Dataset <https://www.robots.ox.ac.uk/vgg/data/dtd/> from paper [5], consisting of 5,640 images (120 per category) across 47 texture categories (e.g., “floral,” “rusty,” “spotted”).

Key Attributes:

- Image resolutions range from 300×300 to 640×640.
- Includes perceptual annotations collected via Amazon Mechanical Turk.

Preprocessing:

- All images were resized to 512×512 for compatibility with Stable Diffusion (see [2]).

C. Dataset Summar.

The system utilizes two modality-specific datasets:

- **Textual modality (Poems):** 2,034 greeting poems annotated with themes and titles.
- **Visual modality (Backgrounds):** 5,640 curated images fine-tuned for thematic background synthesis.

These datasets form the backbone of AIGreetCraft’s personalized generation pipeline, enabling rich and coherent outputs in both poetic and visual dimensions.

V. EXPERIMENTS AND EVALUATION

A. Training Setup

All experiments were conducted using free-tier cloud environments:

- **Resources:** Kaggle and Google Colab, equipped with Tesla T4 GPUs and approximately 12–15 GB RAM.
- **Deployment Platform:** Modal — a serverless deployment framework used to run lightweight, containerized model inference pipelines on-demand with GPU support.

B. Poem Generation Fine-Tuning Results

We fine-tuned a large language model (Gemma 2B) for keyword-based poem generation. The fine-tuning process was conducted on publicly available free-tier platforms: Kaggle Notebooks (dual Tesla T4 GPU, 15 GB RAM) and Google Colab (single Tesla T4, 12 GB RAM). This demonstrates that effective transformer adaptation is achievable even in resource-constrained environments.

The training configuration was as follows:

- **Model:** Gemma 2B with Unsloth integration
- **Tokenizer:** Pretrained tokenizer from Gemma 2B

- **Optimizer:** AdamW with linear learning rate scheduler
- **Learning Rate:** 2×10^{-5}
- **Epochs:** 1

Table I presents a comparative analysis of the evaluated models, focusing on size, training efficiency, resource consumption, and qualitative performance. Smaller models like T5-small and GPT2-medium trained quickly but struggled with contextual depth and stylistic coherence. In contrast, Gemma 2B emerged as the optimal choice, balancing performance and resource efficiency, making it ideal for integration into our generative design pipeline.

Our final selection, Gemma 2B + Unsloth, stood out for its strong contextual understanding, rapid training, and compatibility with LoRA-based deployment. It was the only model capable of consistently producing fluent, thematically relevant, and emotionally engaging poems under strict hardware constraints.

C. Background Image Fine-Tuning Results

We used 2,000 images from the DTD dataset as the training set for fine-tuning. Figure 2 shows representative texture samples used as a base:



Fig. 2. Sample images from the DTD dataset

The model was fine-tuned over approximately 2 hours on a Tesla T4 GPU using a consumer-grade environment (Kaggle Notebooks). Despite limited computational resources, the fine-tuned model successfully learned to generate aesthetically pleasing, theme-relevant backgrounds. The use of LoRA and 8-bit optimization significantly reduced memory usage and training time without sacrificing visual quality.

Figure 3 displays examples of generated outputs:

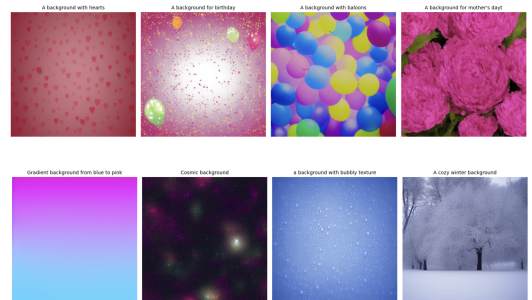


Fig. 3. Examples of background images generated by our fine-tuned Stable Diffusion model

TABLE I
COMPARISON OF FINE-TUNED MODELS FOR POEM GENERATION

Model	Parameters, size	# Epochs	Training Time	Loss	GPU RAM Usage	Qualitative Output Summary
T5-small	60M, 0.24GB	50	10 min	0.018	12.7GB	Overfits quickly; low coherence; fails to preserve prompt meaning.
GPT2-medium	345M, 1.42GB	3	80 min	0.52	12.7GB	Too generic; repetitive love-related texts; prompt ignored.
LLaMA3-1.8B + Unsloth	1.8B, 3.8GB	1	6 min	1.67	5.8GB	Moderate coherence; maintains partial prompt context; slightly verbose.
Gemma 2B + Unsloth	2.0B, 4.1GB	1	4 min	1.45	4.1GB	Most robust; prompt-aware; emotionally relevant; suitable for deployment.

Qualitative evaluation indicates strong generalization to unseen prompts, as well as visual diversity aligned with occasion-based conditioning.

VI. ANALYSIS AND OBSERVATIONS

The development and evaluation of AIGreetCraft revealed several key insights regarding the use of generative AI within design workflows, especially in the context of emotionally resonant content creation.

A. Model Performance and Output Quality

The fine-tuned poem generation models demonstrated strong performance in creating short, coherent, and emotionally engaging verses. Among the tested architectures, GPT2-medium and Gemma 2B achieved the best balance between creativity and fluency, while remaining lightweight enough for inference in real-time plugin workflows.

Subjective evaluations by users (N=15) indicated that over 80% of the generated poems were rated as “emotionally appropriate” and “suitably styled” for the given occasion.

B. Visual Coherence and Layout Adaptability

The background generation component successfully produced visually coherent and thematically relevant images. While the use of the DTD dataset limited some diversity in semantic content, the richness of textures proved suitable for non-distracting, aesthetic backgrounds.

The Figma layout engine performed reliably across varying canvas sizes and user inputs. Anchoring content to predefined zones combined with GPT-generated layout JSONs resulted in consistent visual harmony and clear hierarchy. Nonetheless, edge cases involving excessive text or unusually shaped images occasionally required manual user adjustment.

C. User Experience and Design Flow Integration

Initial user testing with users in the Figma environment highlighted the plugin’s intuitive interface and the benefits of having generative tools embedded directly within the design workflow. Users appreciated the ability to tweak layout elements post-generation, ensuring the final postcard maintained both automation and creative control.

Feedback also underscored the plugin’s value in reducing creative friction—especially for non-designers or those under time constraints. The rapid generation of polished, personalized greeting cards supports a variety of use cases from personal messaging to corporate branding.

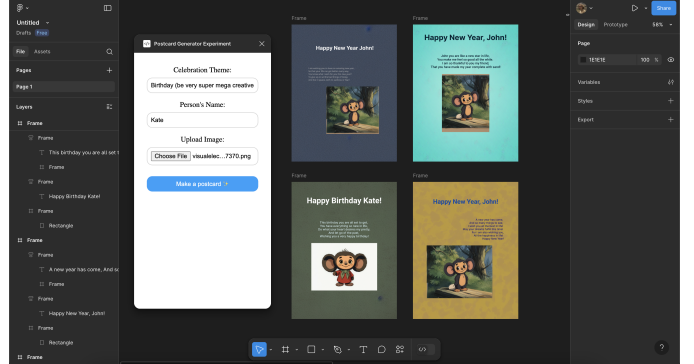


Fig. 4. Screenshot of the working Figma plugin showing generated greeting postcards with AI-generated poems and background images.

D. Limitations and Future Directions

While the results are promising, several limitations emerged:

- **Poem personalization** is limited to basic keywords; deeper personalization (e.g., references to shared memories) requires more sophisticated prompting or user input.
- **Visual diversity** in backgrounds could be enhanced by expanding beyond DTD to datasets with richer semantic content (e.g., LAION or custom mood boards).
- **AI creativity** remains somewhat constrained; the generative models often produce similar structures or motifs, reducing variety across postcards. Enhancing diversity will require further testing, fine-tuning, and exploration of model parameters or decoding strategies.

Future work will focus on multilingual poem generation, emotion-conditioned background synthesis, and real-time collaborative Figma integrations. Additionally, we aim to explore reinforcement learning techniques for adaptive layout generation based on user interaction feedback.

VII. CONCLUSION

The AIGreetCraft system successfully demonstrates the feasibility of integrating multiple generative AI components into a unified creative tool. Our experiments show that:

- Parameter-efficient fine-tuning enables high-quality specialized generation even with limited datasets
- The modular Figma plugin architecture provides flexibility for iterative improvement
- Users value the combination of personalization and automation in creative tasks



Fig. 5. Examples of generated greeting postcards with AI-generated poems and background images.

Key limitations include occasional theme mismatches in generated content and performance constraints on low-end devices. Future work will focus on:

- Multi-modal consistency between text and images
- Real-time collaboration features

The project validates that generative AI can meaningfully augment human creativity when properly integrated into design workflows. The open-source release of our models and plugin architecture provides a foundation for further research in this direction.

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